

Research Design – by J. Creswell

Three bullet points:

- Plan first. Plan again. Plan some more.
- Research design—including conscious choices as to worldview, strategy, and method—is the key part of planning.
- Plot and write the results of your research in a methodical, structured way for best results.

Who is the writer? Who is the audience?

- John W. Creswell
 - Professor, Department of Educational Psychology, University of Nebraska-Lincoln (1978-present)
 - Ph.D., Higher Education, University of Iowa
 - M.A., Student Personnel and Counseling, University of Iowa
 - B.A., History/Political Science, Muskingum College
 - Fulbright Scholar
 - Author of 11 books
 - Editorial Board member or reviewer for Journal of Higher Education, Review of Higher Education, ASHE/ERIC Research Reports, Research in Higher Education, Higher Education Reports
 - all for 20+ years.
- Audience: Primarily graduate students beginning theses or other research projects

Research design

- Comprehensive plan and procedure for assessing a research problem
- Researcher has a choice of designs, each of which will then contain worldviews, strategies, and methods.
- Factors affecting the choice are the research problem itself, the background and interests of the researcher, and the intended audience.

Research Design Categories

- **Quantitative:** numbers, closed-ended, data hypotheses, experiments, deductive
 - A means for testing objective theories by measurement of variables
 - Use when research problem calls for understanding of causality/influence, results of intervention, prediction of outcomes.
- **Qualitative:** words, open-ended, interviews, ethnography, inductive
 - A means for exploring meaning ascribed to social or human problems
 - Use when problem is not well-understood and requires exploration
- **Mixed** – can be combination of Quantitative and Qualitative in parallel, series, or transformational combination
 - Use when problem can not be accurately assessed using only one design.

Worldviews/Philosophies

- Where is the researcher coming from?
- What assumptions does s/he make? (The answer is never “none.”)
- Major Worldviews:
 - (Post)positivism/Determinism/Empiricism
 - Constructivism:
 - Advocacy:
 - Pragmatic:

Worldviews/Philosophies (cont.)

- (Post)positivism/Determinism/Empiricism:
 - Cause and effect/Scientific method
 - Experimentation and measurement can reveal objective reality
 - Strive for objectivity
 - Start with hypothesis and test
- Constructivism:
 - Reality is subjective and multiple
 - Seek to understand meaning as experienced by participants
 - Meaning of events is mediated by interactions with others, and social and cultural context
 - Start with open-ended inquiry and result in hypothesis

Worldviews/Philosophies (cont.)

- Advocacy:
 - Start with point of view as lens to interpret events
 - Focus on marginalized groups and social justice; most participatory and collaborative worldview
 - A deliberate counterweight to more prevalent worldviews
 - Seeks to change reality and not just describe or predict it
 - Key types: Feminist perspectives, queer theory, race and origin discourses, critical theory, disability theory.
- Pragmatic:
 - Analogue of mixed-method strategy
 - Mixes aspects of other worldviews to get at “what works”
 - No need for ideological consistency
 - Believe in objective reality (as with empiricists), but acknowledge context is inescapable (as per constructivists)

Research Strategies

- Specific approaches to inquiry within design categories
- **Quantitative:** Experimental and quasi-experimental, surveys
- **Qualitative:** Narrative, case study, ethnography, grounded theory
- **Mixed:** Parts of both, in parallel, series, or combination

Research Methods

- How to collect, analyze, and interpret data. As with strategies, tend to be associated with specific design categories.
- **Quantitative:** Closed-ended, pre-determined questions, observational or instrumented numeric data, statistical analysis/interpretation
- **Qualitative:** Open-ended, pre-determined or evolving questions, non-numeric data
- **Mixed:** Some of each

Literature Reviews

- Part of the planning stage of the research design – helps show where proposed research problem fits into existing knowledge, why it is worthy of study, and whether contemplated design is possible
- Purposes:
 - Summarize the state of knowledge about the research problem and closely-related areas.
 - Frame the research problem by illustrating gaps in knowledge or conflicting/ambiguous results
 - Place results and conclusions of the research in context by showing where they differ or what they add

Literature Reviews (cont)

- Steps to a literature review:
 - Identify key words and ideas related to the research problem
 - Locate most prominent journal articles and books
 - Use cites from these works to identify other works, and relationships between works
 - Summarize/abstract most relevant works
 - Map key concepts and schools of thought
 - Identify and define key terms/constructs to be used in research design
 - Determine key theories which can be used to explore or explain the research problem

Theories in research

- Theories are hypotheses which specify or explain the relationship among variables
- The “rainbow bridge” between independent (cause or predictor) and dependent (outcome or effect) variables.
- Theories can be micro/meso/macro level
- **Quantitative** research designs typically begin with a theory, and work to disprove or support it
- **Qualitative** research designs may begin with a theory, or build one up inductively from data collected and analyzed.

Ethical considerations in research

- Anticipating and considering ethical issues is a core part of research design
- Is the study properly disclosed to participants?
- Do they give informed consent to participate, and understand that they can leave at any time?
- Does the study risk harm to (or withhold care from) some of the participants?
- Does the research design safeguard participant data and identity?
- Is the language of the write-up inclusive and sensitive to the wishes of participants?
- Does the research document fully credit all those (and only those) actually involved in its design and execution?

Cases, Numbers, Models – by Sprinz & Wolinsky

Three bullet points:

- Methods in scholarly research in international relations are changing, but the literature dealing with IR methods is still outdated
- The field is increasingly quantitative and mode-oriented rather than descriptive
- Quantitative and hybrid analysis in IR is a positive thing for the field, and should be encouraged and supported

Who are the writers? Who is the audience?

- Detlef Sprinz
 - Professor, Department of Economic and Social Sciences, The University of Potsdam
 - Senior Research Fellow I, Center for International Climate and Environmental Research – Oslo
 - Previously Chairman/Member, Scientific Committee, European Environment Agency (2004-2012)
 - Ph.D./M.A (Political Science), University of Michigan. Dissertation: “Why Countries Support International Environmental Agreements: The Regulation of Acid Rain in Europe”
 - M.A. (Economics), University of the Saarland
 - Journals:
 - Global Environmental Politics (Editorial Board)
 - Journal of Environment & Development (Editorial Board)
 - Weather, Climate, and Society (Associate Editor for Political Science)
 - Self-description: “His areas of research and publications encompass long-term policy, inter/national institutions & the evaluation of their performance, European & international environmental policy, and modeling political decisions.”
- Yael Wolinsky-Nahmias
 - Senior Lecturer, Department of Political Science, Northwestern University
 - Co-director, Northwestern Environmental Field School
 - Director, Program In Environmental Policy and Culture
 - Ph.D. Political Science, University of Chicago
 - M.A. Political Science, University of Pennsylvania
 - B.A. International Relations and East Asian Studies, Cum Laude, Hebrew University of Jerusalem
 - Field of active study appears to be environmental policy and its international relations aspects.
- Intended Audience: Practitioners of Research in International Relations

What are IR studies?

- They explain politico-military interactions among countries, societies, and organizations.
- Purpose of the book:
 - Historically IR field is descriptive and qualitative – this is changing
 - More quantitative analysis – use of game theory, statistical methods
 - New sub-fields and more specialization (ex: int'l environmental politics)
 - Existing works on research methodologies in the field have not kept up – this work helps fill the void
 - Advances positivist agenda in field

Findings and Conclusions

- Main Points:
 - No one methodology or perspective is suitable for the entire field
 - IR research problems are susceptible to the use of quantitative analysis, case study method, and other formal models (such as game theory)
 - New methodologies can expand the range of testable (and hence usable) theories in the field
 - Analogy is to political science field, which moved in the quantitative direction from the 1970s
 - Hybrid research designs can add substantial value to the field by increasing reliability of results, but are not yet popular
- Research Design and Findings:
 - Classification of articles published in 6 major journals in the field between 1975 and 2000 into categories based on methodology(ies) employed
 - Number of articles with no methodology has fallen from appx 1/2 to under 1/3 since the late 1970s
 - Greatest increase is in articles using statistical and formal model methodologies

Questions - Creswell

- Research Design
 - Who is the intended audience of your research project?
 - Identify what your worldview is.
 - What factors from your personal/professional background inform your worldview?
 - Identify the research method that is most appealing and state why.
- Literature Reviews
 - Draft and share a central research question.
- Ethical Considerations in Research
 - Identify two anticipated ethical issues that could arise, or that should be considered when constructing your research proposal.

Questions - Sprinz & Wolinsky

- Does the design of the Sprinz & Wolinsky study support their conclusions?
- What biases and pre-conceived opinions might they have about the subject?
- What issues are raised by these potential biases in terms of the conduct of the research study?